

EMPIRICAL IMPLEMENTATION AND OUT-OF-SAMPLE RESULTS

Direct Forecasting of Tail Risks in Brent Crude Oil

Empirical section continuing the methodology of Sections 2.1–2.4.

1 Out-of-Sample Design

1.1 Direct multi-step forecasting

The goal of this part of the analysis is genuinely predictive: we want to anticipate the conditional distribution of Brent returns several periods into the future, using only the information available today. There are two classical ways to produce such multi-step forecasts. The iterated (recursive) approach fits a single one-step model and chains it forward, so that a two-step forecast is obtained by feeding the one-step forecast back into the same equation. This is elegant under Gaussian assumptions but fragile in practice: any misspecification in the one-step model compounds as the horizon grows, and, more decisively for us, a quantile cannot be iterated at all, because the quantile of an expectation is not the expectation of a quantile.

We therefore adopt direct forecasting. Instead of chaining a one-step model forward, we estimate a separate, bespoke quantile regression for each horizon h by shifting the target back by h periods, so that the information available at t is paired with the realised future value Y_{t+h} :

$$\hat{\beta}^{(h)}(\tau) = \arg \min_{\beta} \sum_{t=1}^{T-h} \rho_{\tau}(Y_{t+h} - X_t \beta), \quad Q_{\tau}(Y_{t+h} | X_t) = X_t^{\top} \beta^{(h)}(\tau). \quad (1)$$

Because a dedicated model is fitted at each horizon, the estimator never has to iterate forward and is immune to the compounding bias of the recursive scheme; it simply targets the horizon we care about directly. Crucially, this design lets us study how the geopolitical risk premium propagates over time by re-estimating (1) over a grid of horizons h and watching how the conditional quantiles, and their predictive accuracy, evolve as h increases.

1.2 Rolling-window backtesting

To judge predictive ability honestly, the information used to estimate the model must be kept strictly apart from the information used to score it. We hold out the most recent portion of the sample as a test set, beginning in January 2025 (and extended back to 2023 for the coverage backtests, so that the post-pandemic crisis episodes enter the evaluation), and we re-estimate the model on a fixed rolling window of 250 trading days, roughly one trading year. At each forecast origin the window is fitted, an h -step-ahead forecast of Y_{t+h} is issued, and the window then advances by one day, dropping its oldest observation and adding the newest.

The economic rationale for this design is twofold. First, a tail-risk model is only useful if it works on data it has never seen, so all reported losses and coverage statistics are computed exclusively on the held-out test set. Second, a single forecast is never conclusive: one h -step-ahead prediction could be right or wrong by luck. By rolling the window forward one day at a time we generate a long sequence of overlapping h -step-ahead forecasts, and averaging the loss over

this sequence yields a statistically meaningful measure of predictive capability rather than the artefact of a single origin. The fixed 250-day window also lets the parameters continuously forget superseded regimes and re-anchor on the prevailing one, which matters in a market whose volatility structure is reshaped by every new geopolitical episode.

2 Competing Specifications: Capturing Geopolitical Tail Risk

The standard quantile regression is an excellent description of the typical conditional quantile, but it is, in a precise sense, memoryless about its own failures. It maps today’s geopolitical and financial covariates into a quantile forecast and then forgets whether that forecast was breached. In tranquil markets this is harmless; in the tails it is not.

The economic object we care about is not the centre of the distribution but its extreme edge, the threshold beyond which an oil position suffers a catastrophic loss. And extreme oil moves are not independent draws: they cluster. A supply shock threatening the Strait of Hormuz, or a demand collapse of the COVID-19 type, does not resolve in a single session; it propagates through inventory decisions, futures-curve repositioning and risk re-pricing over days and weeks. Empirically this shows up as volatility clustering in the tail, where a Value-at-Risk breach today materially raises the probability of another breach tomorrow. A model that treats each tail event as an isolated surprise is therefore caught off guard precisely when accuracy matters most.

This is the gap that Engle and Manganelli (2004) close with the CAViaR framework, in which the conditional quantile is allowed to depend autoregressively on its own past behaviour, so that the tail remembers recent stress. We do not estimate a full CAViaR recursion; instead we import its central economic insight into the direct-forecasting design by augmenting the standard specification with variables that encode the model’s own recent boundary violations. The purpose is singular: to let the forecast adapt to clustered geopolitical stress and thereby behave better in the tails. Two such augmentations are considered, differing only in how much they tell the model about a breach.

2.1 Model 1: standard quantile-regression direct forecast

The benchmark is the smoothed-absolute-dynamics specification, projected directly onto the future:

$$Q_\tau(Y_{t+h} \mid \Omega_t) = \alpha_\tau + \beta_\tau \text{GPRD_MA7}_t + \gamma_\tau^\top X_t. \quad (2)$$

It is parsimonious and serves as the yardstick against which any added tail machinery must justify itself.

2.2 Model 2: CAViaR binary-indicator augmentation

The first augmentation tells the model whether it was recently wrong. For every date t we compare the realised value y_t against the quantile boundary that the model had projected h periods earlier, and raise two asymmetric flags, one for the upper tail and one for the lower, so that supply-driven surges and demand-driven collapses can feed forward through separate channels:

$$\text{upside_breach}_t = \mathbf{1}\{y_t > Q_{\text{high}}(y_t \mid x_{t-h})\}, \quad (3)$$

$$\text{downside_breach}_t = \mathbf{1}\{y_t < Q_{\text{low}}(y_t \mid x_{t-h})\}. \quad (4)$$

The asymmetry is economically deliberate: an upper-tail breach signals a supply-side panic, a lower-tail breach a demand-side rout, and the two are allowed to shift the forecast in opposite directions. The boundaries are fitted at the $\tau = 1\%$ and 99% quantiles; by comparing y_t only with a boundary computed h periods earlier, the flags are constructed without any look-ahead. Over the full sample the construction yields 40 upside and 36 downside breaches against the $1\%/99\%$ envelope, a sparse but pointed signal of regime stress.

2.3 Model 3: CAViaR severity augmentation

A binary flag records that a crash happened but is blind to its violence: a 1% breach and a 10% breach speak with the same voice. Yet for tail risk the magnitude is the whole story, the difference between a manageable loss and a blow-up. The severity variant therefore replaces the indicators with the absolute size of the exceedance, the fallout error:

$$\text{upside_severity}_t = \max(0, y_t - Q_{\text{high}}(y_t | x_{t-h})) \geq 0, \quad (5)$$

$$\text{downside_severity}_t = \max(0, Q_{\text{low}}(y_t | x_{t-h}) - y_t) \geq 0. \quad (6)$$

Entering the fallout error in absolute value forces its coefficient to act as a pure stress-intensity loading, uncontaminated by the sign of the exceedance, so the upper- and lower-tail severities read as independent measures of how hard the market overshot. Because both augmentations need a realised value to define a breach, the stress signal they carry is sharp at very short horizons but decays as h lengthens, an economic limitation the out-of-sample results will make concrete.

3 Evaluating the Forecasts

Three complementary lenses judge the forecasts: a strictly proper loss measured over the rolling window, a magnitude-of-exceedance audit, and a pair of formal coverage backtests. The mechanics of every test are derived in full in the companion Annex on Out-of-Sample Forecast Evaluation Tests; here we keep the treatment brief and focus on what each test certifies.

3.1 The pinball loss and the generalization gap

A quantile cannot be scored with the symmetric mean-squared error, because over- and under-prediction must be penalised by different amounts. The asymmetric pinball (check) loss supplies that asymmetry:

$$L_\tau(e) = e \cdot (\tau - \mathbf{1}\{e < 0\}), \quad e = Y_{t+h} - \hat{y}_{t+h|t}(\tau). \quad (7)$$

It is the only strictly proper scoring rule for quantiles: it is minimised if and only if the forecast equals the true conditional quantile, so a lower out-of-sample pinball loss is conclusive evidence of a better tail estimate. Tracking the loss on the training data against the loss on the test data across a grid of horizons yields the generalization gap; a gap that widens rapidly with h is the hallmark of overfitting. Averaging the loss over the whole rolling window for each (h, τ) pair, the loss-minimising horizon per model and quantile is reported in Table 1.

Model	$\tau = 0.05$ (best h , loss)	$\tau = 0.50$ (best h , loss)	$\tau = 0.95$ (best h , loss)
Standard QReg	$h=7$ (0.5295)	$h=3$ (1.6061)	$h=1$ (0.5624)
CAViaR-indicator	$h=9$ (0.6342)	$h=1$ (1.6210)	$h=1$ (0.5206)
CAViaR-severity	$h=5$ (0.6860)	$h=1$ (1.6457)	$h=1$ (0.5432)

Table 1: Rolling out-of-sample pinball loss: horizon attaining the lowest average loss per model and quantile (lower is better). The horizons are not selected as optima; they are reported across the grid that we evaluate deliberately.

The CAViaR variants attain the lowest upper-tail loss ($\tau = 0.95$) but only at the immediate horizon $h = 1$, where the breach signal is freshest; the benchmark holds up across the median and the lower tail at the longer horizons. This is the empirical fingerprint of the stale-information limitation flagged in Section 2.

3.2 Fallout (severeness) audit

Coverage frequency alone can hide catastrophic single-day failures, so we also audit the magnitude of every breach: how far the realised return overshot the predicted 1%–99% tunnel. For risk management, frequent but tiny breaches are far preferable to rare but massive ones, since the latter betray a structural failure to price an impending shock. At $h = 2$ the three models keep the realised return inside the predicted envelope on 97.86% (Standard), 97.26% (CAViaR-indicator) and 97.26% (CAViaR-severity) of test days, with no isolated catastrophic overshoot. The full breach paths for the three models, including the behaviour of the tunnel during the 2020 crisis, are collected in the Graphical Annex (Figures G.1–G.3) and discussed in Section 4.

3.3 Coverage backtests: from Kupiec to Christoffersen

Because the model’s output is an estimated quantile that functions as a Value-at-Risk threshold, its adequacy is judged with risk-management backtests built on the binary hit sequence I_{t+h} , which equals 1 when the realised return breaches the forecast and 0 otherwise.

The natural first question is one of frequency, answered by the Kupiec (1995) Proportion-of-Failures test. If the model is correctly calibrated, the τ -quantile should be breached a fraction τ of the time, so the test compares the empirical hit rate $\hat{\pi} = x/T$ against the nominal τ through a likelihood ratio that is asymptotically χ_1^2 . Passing it confirms that the model breaches at the right rate on average. But frequency is only half the story: Kupiec is blind to timing. A model could breach exactly $\tau\%$ of the time yet concentrate all those violations inside a single turbulent fortnight, catastrophic behaviour that an average rate conceals. This is the deficiency the Christoffersen test repairs, and the one we develop in detail.

The Christoffersen combined (conditional coverage) test

Christoffersen (1998) asks not only how many breaches occur but when they occur, by modelling the hit sequence as a first-order Markov chain. Counting the four transition types $T_{00}, T_{01}, T_{10}, T_{11}$, it estimates the probability of a breach today conditional on yesterday’s state,

$$\pi_{01} = \frac{T_{01}}{T_{00} + T_{01}}, \quad \pi_{11} = \frac{T_{11}}{T_{10} + T_{11}}. \quad (8)$$

If the model has truly absorbed geopolitical shocks, the fact that oil crashed yesterday should already be priced into today’s forecast, so a breach today must be independent of a breach yesterday. The independence hypothesis $H_0 : \pi_{01} = \pi_{11} = \pi$ is tested by the likelihood ratio

$$LR_{IND} = -2 \ln \frac{L_{\text{indep}}}{L_{\text{dep}}} \sim \chi_1^2. \quad (9)$$

Christoffersen’s decisive move is to combine this with Kupiec’s frequency test. Because the two statistics are asymptotically independent, their sum is again chi-squared, with degrees of freedom adding:

$$LR_{CC} = LR_{UC} + LR_{IND} \sim \chi_2^2. \quad (10)$$

A single statistic thus certifies both properties at once: the forecast breaches at the right rate and at independent times. This is the most demanding coverage criterion and the one we use to validate the production model. For the benchmark at $\tau = 0.95$, $h = 2$, over the 867-day window from 2 January 2023 to 28 April 2026, the combined test returns $p = 0.136$; the CAViaR variants return an even more decisive $p = 0.445$. None of the three models exhibits breach clustering, and all pass at the 5% level.

3.4 Diebold–Mariano comparison of the three models

The coverage tests certify each model in isolation; the Diebold–Mariano (1995) test arbitrates between them. Using the asymmetric tick loss and a HAC variance with bandwidth $h - 1$, which matches the $MA(h - 1)$ autocorrelation of h -step direct forecast errors, it tests the null of equal predictive accuracy, where a negative significant α favours the first model. At $\tau = 0.95$, $h = 2$:

Comparison	α	t -stat	p -value	Verdict
Standard vs CAViaR-i	−0.00647	−0.874	0.382	no sig. difference
Standard vs CAViaR-s	−0.00620	−0.925	0.355	no sig. difference
CAViaR-i vs CAViaR-s	+0.00027	+0.158	0.875	no sig. difference

Table 2: Pairwise Diebold–Mariano tests, tick loss, $\tau = 0.95$, $h = 2$.

The companion error metrics confirm a statistical dead heat: average tick losses of 0.2516 (Standard), 0.2581 (CAViaR-i) and 0.2578 (CAViaR-s). No comparison is significant even at the 10% level, so on the pinball/tick-loss criterion the three models are statistically indistinguishable in predictive accuracy at the operative horizon.

4 Model Selection and Forecast Horizons

4.1 Forecast horizons

We do not search for a single optimal horizon. The aim is precisely the opposite: to map how geopolitical tail risk propagates over time, so the model is evaluated deliberately across a grid of horizons. In reporting results we privilege horizons that correspond to economically meaningful short and medium periods, an immediate-shock window of a couple of trading days and a medium window of a few trading weeks to a month, since these are the windows over which a supply or demand shock transmits into physical and derivative oil markets. The choice of which horizons to display is informed by this economic reading, not by optimising a loss.

4.2 Choice of specification

On the headline predictive-accuracy criterion the three models tie: the Diebold–Mariano test finds no significant difference in tick loss at the operative horizon, and the CAViaR augmentations sharpen the upper tail only marginally and only at $h = 1$. Taken alone, this would argue for the parsimonious benchmark.

The comparison changes, however, once we look beyond a single horizon and inspect breach behaviour across the entire test period. To do this we use the breach indicators, which let us evaluate the realised quantile bounds against the data over the whole sample rather than at one (h, τ) point. The picture that emerges, shown in the Graphical Annex (Figures G.1–G.3), is clearly favourable to the CAViaR specifications. The standard model controls its breaches only by inflating its tails violently during crises: around the 2020 collapse its 1% and 99% bounds explode toward roughly +600 and −200, and it still accumulates 69 breaches over the sample at an average exceedance severity of 2.47. The two CAViaR models, by contrast, cut the breach count to 29 and the average severity to about 1.85, while their crisis tunnel widens far less, peaking near +300.

The economic reading is exactly the one that motivated the augmentation. Feeding the model its own recent violations lets the tail react to clustered stress in time: when a breach occurs, the conditional quantile adjusts on the next step, so the model retains its breaches without having to over-expand the entire envelope. The standard model, lacking this memory, can only stay calibrated by widening its tails indiscriminately whenever volatility spikes, which is both less informative and operationally costly for risk management.

Because the overarching objective of this thesis is to forecast the full conditional distribution of Brent returns across a span of horizons, we retain the standard quantile-regression direct forecast as the baseline production model: it is parsimonious, passes every coverage backtest, and is statistically tied on tick loss. We nonetheless carry the CAViaR specifications forward as the tail-robust complement, since across the whole test period they deliver fewer and milder breaches with markedly better-behaved crisis tails, the property that matters most when the model is used to price extreme geopolitical risk.

The complete derivation, hypotheses, test statistics and per-model pass/fail evidence for the pinball, generalization-gap, fallout, Kupiec, Christoffersen-independence, Christoffersen-combined and Diebold–Mariano tests are collected in the test Annex; the breach paths of the three models are in the Graphical Annex.